

Brock Biology of Microorganisms, 15e (Madigan et al.)
Chapter 16 Diversity of Bacteria

16.1 Multiple Choice Questions

1) Which phylum harbors the most functional diversity observed and also has the most cultured representatives?

- A) *Actinobacteria*
- B) *Bacteroidetes*
- C) *Firmicutes*
- D) *Proteobacteria*

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.1

2) Isolating a bacterium of the _____ class would be of high impact due to it only having one cultured representative.

- A) *Alphaproteobacteria*
- B) *Betaproteobacteria*
- C) *Epsilonproteobacteria*
- D) *Zetaproteobacteria*

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.5

3) A pink-pigmented facultative methylotrophic bacterium is LEAST likely to be classified as a member of

- A) *Rhizobiales*.
- B) *Rhodobacter*.
- C) *Methylobacterium*.
- D) *Proteobacteria*.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.1

4) Which genus would you most likely find in the photic layer of the oceans due to its abundance and aerobic chemoorganotrophic phenotype?

- A) *Acetobacter*
- B) *Nitrosomonas*
- C) *Paracoccus*
- D) *Pelagibacter*

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.1

5) Which of the following traits is NOT a characteristic of *Caulobacter*?

- A) abundant in aquatic habitats rich in nutrients
- B) produces a cytoplasmic outgrowth (prostheca)
- C) aerobic chemoorganotrophs
- D) asymmetric cell division

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.1

6) Treating wastewater requires the removal of organic material, and an ideal phenotype of a bacterium to do this would flocculate to make the cells settle (rather than staying in suspension) in a wastewater stream. Which species would be best for this application?

- A) *Bartonella quintana*
- B) *Burkholderia cepacia*
- C) *Wolbachia pipientis*
- D) *Zoogloea ramigera*

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.2

7) While not all members of _____ are pathogenic, two medically relevant species cause meningitis and gonorrhea in humans.

- A) *Enterobacter*
- B) *Neisseria*
- C) *Shigella*.
- D) *Staphylococcus*

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.2

8) Chemolithotrophic ammonia-oxidization is a metabolic requirement for members of

- A) *Nitrosomonas*.
- B) *Nitropseudomonas*.
- C) *Nitrosomonas* and *Nitrospira*.
- D) *Nitrosomonas*, *Nitrospira*, and *Nitropseudomonas*.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.2

9) Biosynthesis of vitamin K (menaquinone) by _____ spp. inhabiting the _____ of humans is a potentially important mutualistic relationship for human health.

- A) *Bifidobacterium* / intestinal tract
- B) *Bifidobacterium* / stomach
- C) *Escherichia* / intestinal tract
- D) *Escherichia* / stomach

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.3

10) A gram-negative, obligately aerobic, bacillus-shaped bacterium that can degrade a large variety of aromatic compounds (making them potentially useful for bioremediation) is likely a(n)

- A) actinobacterium.
- B) enteric.
- C) pseudomonad.
- D) sphingomonad.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.1

11) Which characteristic is NOT associated with *Vibrio* spp.?

- A) aquatic
- B) gliding motility
- C) gram-negative
- D) pathogenic

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.4

12) A pseudomonad that loses its R plasmid in the soil is LEAST likely to

- A) cause an infection in a human.
- B) colonize a surface.
- C) infect a shrub.
- D) survive a habitat dominated by antibiotic-producing *Streptomyces* spp.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.4

13) Which of the following is NOT a member of the *Deltaproteobacteria*?

- A) *Campylobacter*
- B) *Desulfovibrio*
- C) *Geobacter*
- D) *Syntrophobacter*

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.5

14) Which of the following is NOT a trait of rickettsias?

- A) They are obligate intracellular parasites.
- B) They lack cell walls.
- C) They have not been cultivated in the absence of host cells.
- D) Both the host and parasite are required in order to be alive and metabolically active.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.1

15) Which of the following best distinguishes deltaproteobacteria from epsilonproteobacteria?

- A) Deltaproteobacteria are most commonly found at anoxic-oxic interfaces while epsilonbacteria are most commonly found in anoxic environments.
- B) Many epsilonbacteria oxidize sulfur compounds, which could be produced by a sulfur-reducing deltaproteobacterium.
- C) A sulfate-reducing epsilonbacterium produces sulfite, which a deltaproteobacterium reduces further to hydrogen sulfide.
- D) An epsilonbacterium reduces sulfur compounds whereas all deltaproteobacteria oxidize sulfur compounds.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.5

16) Which of the following organisms is most likely to form a relationship with *Syntrophobacter wolinii*?

- A) a bacterium that oxidizes acetate
- B) a bacterium that consumes sulfate
- C) a bacterium that consumes H₂
- D) a bacterium that consumes pyruvate

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.5

17) Based on oxygen requirements and metabolic strategies, you would most likely isolate an epsilonproteobacterium from

- A) anoxic and sulfate-rich waters.
- B) a plant's rhizosphere.
- C) sulfate-rich anoxic-oxic interfaces.
- D) sulfide-rich soils.

Answer: C

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 16.5

18) Which of the following is NOT true of *Epsilonproteobacteria*?

- A) They include the genera *Campylobacter* and *Helicobacter*.
- B) Many species require relatively low O₂ levels to grow.
- C) *Epsilonproteobacteria* are present in freshwater, marine systems, and terrestrial environments.
- D) *Epsilonproteobacteria* are prolific metal reducers.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.5

19) Which taxon is NOT gram-positive?

- A) *Actinobacteria*
- B) *Firmicutes*
- C) *Sphingobacteria*
- D) *Streptococcus*

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.6

20) Which feature(s) differentiate(s) *Actinobacteria* from *Firmicutes*?

- A) absence/presence of a cell wall
- B) endospore production capability
- C) motility and chemotaxis abilities
- D) relative G+C content in their genomes

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.6

21) The order *Lactobacillales* within *Firmicutes* are also called the lactic acid bacteria, because these bacteria

- A) can aerobically catabolize lactic acid.
- B) can ferment lactate.
- C) produce lactate.
- D) uniquely require lactic acid as a precursor for cell wall biosynthesis.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.6

22) What would most likely result if an exponentially growing culture of *Lactobacillus* spp. fermenting glucose was accidentally transferred into a highly aerated environment?

A) Aerobic catabolism of glucose would begin rather than aerobic degradation, due to the presence of O₂.

B) Fermentation of glucose would continue, because the cells are insensitive to O₂ despite lacking aerobic metabolism capabilities.

C) The cells would halt their metabolism due to the presence of N₂, which acts as a global repressor of fermentation in this group.

D) The population would die due to the presence of O₂, because it is a strict anaerobe.

Answer: B

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 16.6

23) Which bacterial genus would be best to select for fermenting cabbage to make homemade sauerkraut?

A) *Clostridium*

B) *Lactobacillus*

C) *Propionibacterium*

D) *Staphylococcus*

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.6

24) Which phenotypic assay could be used to determine whether a new *Streptococcus* spp. is of the pyogenes or viridans group?

A) catalase assay with H₂O₂

B) hemolytic type after growth on a blood agar plate

C) occurrence of cells being in clusters or as individual cells under a microscope

D) product(s) formed during lactose fermentation

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.6

25) *Staphylococcus aureus* is known to cause all of the following diseases EXCEPT

A) acne.

B) arthritis.

C) boils.

D) dental caries.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.6

26) Which species does NOT produce endospores?

- A) *Bacillus cereus*
- B) *Clostridium pasteurianum*
- C) *Listeria monocytogenes*
- D) *Paenibacillus popilliae*

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.7

27) A gram-negative, nonsporulating rod that is highly motile, oxidase negative, facultatively aerobic, ferments glucose, and urease-positive is likely to be of which genus?

- A) *Pseudomonas*
- B) *Proteus*
- C) *Lactobacillus*
- D) *Rickettsia*

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.3

28) A yogurt, which had "contains live cultures" on its label, was transferred onto an acidic carbohydrate-containing medium for selective enrichment. Eventually, isolates of the _____ genus were obtained.

- A) *Lactobacillus*
- B) *Leuconostoc*
- C) *Peptococcus*
- D) *Streptococcus*

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.6

29) A moistened swab rubbed onto a person's forearm then onto a nutrient agar plate with 7.5% NaCl would most likely result in the isolation of _____ spp.

- A) *Clostridium*
- B) *Halobacterium*
- C) *Sarcina*
- D) *Staphylococcus*

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.6

30) Which genus is most easily distinguishable from others below by occurring in cell clusters of 8 or more?

- A) *Peptostreptococcus*
- B) *Sarcina*
- C) *Staphylococcus*
- D) *Streptococcus*

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.7

31) The ability to form endospores in bacteria within *Bacillales*, *Clostridiales*, and *Lactobacillales* suggests the genotype/phenotype evolved

- A) at least three separate times in these divergent orders.
- B) at least once, and horizontal gene transfer among relatives explains its presence in other orders.
- C) only once in an ancestor to these three orders, due to the high number of genes required.
- D) either at least once and horizontal gene transfer among relatives occurred or once in a common ancestor.

Answer: C

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 16.8

32) Which organisms would most likely be isolated from aerobic growth conditions with acetate and a soil sample heated to 80°C for 10 minutes?

- A) aerobic or facultative anaerobic acetogens
- B) endospore-forming bacteria
- C) hyperthermophilic heterotrophic bacteria
- D) mycoplasmas

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.8

33) Which of the following is LEAST commonly associated with endospore-forming bacteria?

- A) They are coccus-shaped organisms.
- B) They are primarily soil dwellers.
- C) They can survive adverse environmental conditions.
- D) They are saprophytic.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.8

34) Which bacillus-shaped genus are primarily intracellular parasites of arthropods such as crustaceans and insects?

- A) *Coxiella*
- B) *Klebsiella*
- C) *Roseospirillum*
- D) *Wolbachia*

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.1

35) *Campylobacter* and *Helicobacter* are two genera within the _____ class of *Proteobacteria*.

- A) *Alphaproteobacteria*
- B) *Betaproteobacteria*
- C) *Deltaproteobacteria*
- D) *Epsilonproteobacteria*

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.5

36) Which group in *Bacteria* lacks cell walls?

- A) *Corynebacterium*
- B) *Mycoplasma*
- C) *Sporosarcina*
- D) *Tenericutes*

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.9

37) Mycoplasmas lack cell walls, thus

- A) they have a larger cytoplasmic volume.
- B) frequently exist as obligate symbionts.
- C) they have a higher sensitivity to osmotic lysis.
- D) they are unable to survive conditions under which protoplasts lyse.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.9

38) The need for additional supplementation of _____ make developing growth media to culture mycoplasmas especially challenging.

- A) unsaturated fatty acids
- B) sterols
- C) vitamins
- D) unsaturated fatty acids, sterols, and vitamins

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.9

- 39) Snapping division is a hallmark trait of _____ which can be visualized under a microscope by _____.
- A) *Actinobacteria* / two adjacent cells linked, appearing as an M dividing into four
 - B) *Actinobacteria* / one cell appearing as a V dividing into two
 - C) Coryneform bacteria / two adjacent cells linked, appearing as an M dividing into four
 - D) Coryneform bacteria / one cell appearing as a V dividing into two

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.10

- 40) Due to the presence of mycolic acids in the cell wall of *Mycobacterium* spp., the _____ staining procedure can be used as a diagnostic tool to identify this genus.

- A) acid-fast
- B) DAPI
- C) Gram
- D) lipo

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.11

- 41) Which bacterial genus is LEAST related to the others listed?

- A) *Actinomyces*
- B) *Mycobacterium*
- C) *Nocardia*
- D) *Streptomyces*

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.12

- 42) A soil sample was diluted into a nutrient broth, then spread onto a plate and grown for one week at room temperature. Several colonies appeared dusty or fuzzy and yet were identified as bacteria—not as fungi. Which genus do these isolates likely belong in?

- A) *Actinomyces*
- B) *Mycobacterium*
- C) *Nocardia*
- D) *Streptomyces*

Answer: D

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 16.12

43) The _____ produce over 500 characterized antibiotics.

- A) *Bacteroidetes*
- B) mycobacteria
- C) pseudomonads
- D) streptomycetes

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.12

44) Members of the _____ are the most abundant bacteria present in the large intestine of humans and contain unusual _____ in their cell membranes.

- A) *Bacteroides* / glycerolipids
- B) *Bacteroides* / sphingolipids
- C) *Sphingobium* / glycerolipids
- D) *Sphingobium* / sphingolipids

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.13

45) Members of the _____ are particularly abundant in the oceans, chemoorganotrophic, rarely pathogenic, and often produce pigments when grown on complex media.

- A) *Clostridiales*
- B) *Flavobacteriales*
- C) *Pelagibacterales*
- D) *Rhizobiales*

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.14

46) Bacteria in the phylum _____ occur as elementary or reticulate bodies depending on their life cycle stage, are obligate intracellular parasites, and also have relatively small genomes.

- A) *Bacteroidetes*
- B) *Chlamydiae*
- C) *Planctomycetes*
- D) *Rickettsia*

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.15

47) Which of the following is NOT a trait of the *Planctomycetes*?

- A) undergo budding cell division
- B) have intracellular compartmentalization
- C) lack peptidoglycan
- D) undergo plant symbiosis

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.16

48) Which structural feature is common in BOTH *Planctomycetes* and *Verrucomicrobia* but RARELY found in bacteria outside these two phyla?

- A) intracellular compartmentalization
- B) prosthecae appendages
- C) sheathlike toga cell envelopes
- D) tubulins

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.16, 16.17

49) Despite ether-linked lipids being a hallmark of the *Archaea*, the thermophilic sulfate-reducing bacteria in _____ also contain ether-linked lipids, which suggests they may be more closely related to *Archaea* than many other groups in *Bacteria*.

- A) *Aquifex*
- B) *Thermodesulfobacterium*
- C) *Thermus*
- D) *Verrucomicrobium*

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.18

16.2 True/False Questions

1) The most metabolically diverse phylum of *Bacteria* known to date is the *Proteobacteria*.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.1

2) *Agrobacterium tumefaciens* can form root nodules in plants and can mate with bacteria, which makes it an excellent host organism for genetic transformations among different domains of life.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.1

3) *Rhodobacter* and *Roseobacter* are two genera with a common ancestor in *Alphaproteobacteria*.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.1

4) The enteric bacteria are a group classified within *Proteobacteria* that cause disease in animals.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.3

5) Due to the genetic similarities of non-pathogenic and pathogenic enterics, phenotypic assays are often performed in favor over taxonomy-based sequencing/probing to distinguish pathogens from non-pathogens.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.3

6) *Betaproteobacteria* is a functionally diverse group that primarily uses anaerobic metabolisms.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.2

7) Many gammaproteobacteria members are obligate intracellular symbionts of eukaryotes, which also enables many to become human pathogens.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.3

8) Pseudomonads are aerobic, gram-positive, polar flagellated, bacillus-shaped cells.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.4

9) Sulfur metabolism is a common trait in *Deltaproteobacteria*.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.5

10) To subgroup the lactic acid bacteria, which are incapable of aerobic degradation and only do fermentation, identifying products formed is a useful way to distinguish homofermentative versus heterofermentative lactic acid bacteria.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.6

11) Many *Listeria* spp. grow well at cold temperatures, so although refrigeration is excellent at preventing growth of most other pathogens, it is NOT very useful at minimizing growth of contaminated foods.

Answer: TRUE

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 16.7

12) Simple staining and microscopy could easily distinguish the different taxa intermixed within a community of *Sarcina*, *Staphylococcus*, and coryneform bacteria.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.10

13) Due to the slow and varied growth rates, mycobacteria are commonly grouped into either slow growing or very slow growing, which has consequences on the growth medium selected for culturing.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.11

14) Except that they are not produced in the cytoplasm, conidia produced by *Streptomyces* spp. are chemically quite similar in composition to endospores produced by *Bacillus* spp.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.12

15) Some *Streptomyces* spp. can biosynthesize several chemically unrelated structures that act as different antibiotics.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.12

16) Chlamydiae are most commonly transmitted throughout environments by insects.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.15

17) When looking under a transmission electron microscope, you observe pirellulosome in a bacterium. This bacterium likely belongs to the *Planctomycetes*.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.16

18) *Thermotoga* species are hyperthermophiles and appear relatively closely related to *Archaea* in comparison to other bacteria such as *Salmonella*.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.18

19) The hypothesis that H₂ was a key electron donor for energy metabolism in primitive organisms is supported by it being such a common trophic strategy in *Aquificae*.

Answer: TRUE

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 16.19

20) *Deinococcus radiodurans* is capable of withstanding especially high levels of radiation, and the carotenoids they produce enhance this resistance.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.20

21) Some species in the genus *Acidobacteria* are classified in *Bacteria* while others are considered *Archaea*.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.21

22) Antibiotics that disrupt or halt cell wall biosynthesis are actually ineffective at harming growth of mycoplasmas.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 16.9

16.3 Essay Questions

1) During stationary phase, group members of *Serratia* produce prodigiosins that contain pyrrole. Considering its structural relatedness to other pyrrole-containing molecules of known function (e.g., chlorophylls, bacteriochlorophylls, porphyrins), propose a potential ecological role for these compounds in soil where they are involved in something other than photosynthesis.

Answer: Answers will vary but should focus on electron transfer mechanisms that pyrrole rings provide to prodigiosins. For example, cells producing them may be able to use electrons from metals in the soil to facilitate oxidative damage to other neighboring cells and establish its niche. The compounds, being either reduced or oxidized, might also be able to tell the cells the type of environmental conditions they are in and coordinate other physiologies accordingly.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 16.3

2) *Proteobacteria* members thrive in soils and occur at high concentrations as well. Other bacterial taxa are considered much less diverse in terms of their metabolic capabilities. Due to the role isolation has in discovering metabolic pathways, consider how obtaining hundreds of new isolates from a less studied phylum might change our views on metabolic diversity. Predict how the results of discovering another highly metabolically diverse group might be handled in terms of taxonomic assignment.

Answer: Regardless of novel metabolisms observed, the broad taxonomic identity of the isolates will not change. Depending on the types of metabolisms, subgroups might be considered whether certain taxa appear unified by a particular function. Analogous to the subgroups of enterics based on differences of fermentation capabilities, this could perhaps result in phenotypic assaying to infer taxonomy.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 16.1

3) Explain how the enteric bacteria can be differentiated by the type and proportion of fermentation products produced by the anaerobic fermentation of glucose.

Answer: While glucose is fermented by all enterics, the two main patterns of products formed are called the 2,3-butanediol fermenters and the mixed acid fermenters. Butanediol, ethanol, CO₂, and H₂ are the primary products of 2,3-butanediol fermentation, and small quantities of acids are also formed. Mixed acid fermentation produces considerably more acetic, lactic, and succinic acids and only small quantities of ethanol, CO₂, and H₂.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.3

4) *Cytophaga columnaris* causes columnaris (or "cottonmouth") disease in fish such as salmon. Based on its taxonomic group, predict some of the activities this bacterium has during infection of fish. Also mention one related taxon that likely has some of the same functions and another taxon that does not.

Answer: Because *C. columnaris* is a member of the *Cytophagales* and also related to the *Flavobacteriales*, it is likely this organism degrades large molecular weight compounds and uses extracellular enzymes to accomplish this task. Its taxonomy also indicates it is an aerobic gram-negative bacillus-shaped bacterium such as other relatives in *Cytophagales* and *Flavobacteriales*. Unrelated taxa with dissimilar functions could range from gram-positives such as *Actinobacteria* and *Firmicutes* to gram-negatives such as *Aquificae* and *Proteobacteria*.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 16.14

5) Even to a well-trained microbiologist, isolates of *Streptomyces* when grown on a nutrient agar plate can appear as though they are fungi. Describe key similarities of how these two unrelated groups can so easily be mistaken for each other, and then propose methods to conclusively distinguish between them.

Answer: To the naked eye, *Streptomyces* appear fuzzy on nutrient agar plates similar to some fungi. Even under a microscope, the hyphal filaments, appearing as mycelial networks of cells, are similar to the morphology of some fungi. Growth of large molecular weight compounds in the nutrient agar is also common in both groups. Due to differences in the domains of life that *Streptomyces* and fungi occur, genetic-based 16S rRNA gene sequencing and 18S rRNA gene sequencing would reveal their proper identities. Staining for the presence of peptidoglycan would also indicate the *Streptomyces* are not eukaryotic.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 16.12

6) Which group of bacteria is most famous for their diversity of antibiotics produced? What could be an ecological role of antibiotic production by bacteria when the antibiotics are not produced at a sufficiently high level to be lethal?

Answer: The *Streptomyces* are well-known for producing hundreds of structurally diverse antibiotics. Even if their production levels in the natural environment are low, they might still serve as stressors to neighboring bacteria. Another possibility is production of certain antibiotics fosters a mutualistic relationship with other microorganisms that can catabolize the substrates. For antibiotics produced during stationary phase, a third possible role could be that they serve as signaling molecules for nutrient availability and/or cell concentration.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 16.12

7) Considering the evolution of *Bacteria*, the criteria used for taxonomy, and the genetic requirements for individual metabolic pathways, explain how it is unlikely to observe three genera of the same family where one genus has strict anaerobes that degrade only small aromatics, another genus degrades large molecular weight sugars aerobically, and a third is a photoautotroph.

Answer: Answers will vary but should emphasize the metabolism of a common ancestor being incompatible with these physiologies. The evolution of such organisms within a taxonomic family might be expected to rarely produce such different metabolisms. The genetic requirements for these metabolisms are quite high and therefore unlikely to evolve quickly within a family.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 16.21

8) Select a group within *Bacteria* that has rare cell morphology observable with a microscope. Describe the appearance, the group that it corresponds to, and another group that would certainly be ruled out when observing this type of cell morphology.

Answer: Answers will vary based on the taxonomic group selected. Possible answers include: *Sarcina* cells occurring in packets of eight or more, coryneform bacteria being V-shaped due to their cell division strategy, *Planctomycetes* stalked cells that also bud, many projecting prosthecae in *Verrucomicrobia*, and intracellular compartments in *Planctomycetes*.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 16.21

9) Compare and contrast the human pathogens and non-pathogens in *Clostridium*. Propose a reason for why this differentiation is challenging to assay in the laboratory (NOT on living eukaryotic animal models).

Answer: Irrespective of their pathogenicity is the unifying anaerobic metabolism of *Clostridium*. Some can ferment large polysaccharides, whereas others only ferment small sugars. What differentiates some as pathogens is not in their ability to ferment certain compounds, which would be easy to assay a *Clostridium* isolate for in the lab. Pathogenic clostridia produce additional compounds that act as toxins to humans. Because each *Clostridium* pathogen produces structurally different toxins, individually identifying all produced chemicals to determine whether each is a toxin is much more challenging.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 16.8